

James Zhang

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EDUCATION

Yale University

Pre-Doctoral Student

Sep 2025 – Dec 2025

GPA: 4.0/4.0

Returning to industry after conducting PhD-level empirical asset pricing research with Professor Bryan Kelly

PhD Coursework: Financial Economics, Econometrics

University of Maryland, College Park

B.S. Computer Science & Mathematics, Minor in Computational Finance

Aug 2022 – May 2025

GPA: 3.9/4.0

Teaching Assistant: Portfolio Management, Computational Finance Capstone

Coursework: Algorithms, Computer Networks, Deep Learning, Probability Theory, Mathematical Statistics, Linear Algebra

PUBLICATIONS

S. Nawathe, R. Panguluri, **J. Zhang**, S. Venkatesh. [Multimodal Deep Reinforcement Learning for Portfolio Optimization](#). *arXiv preprint arXiv:2412.17293*, 2024.

EXPERIENCE

Independent Algorithmic Trading – Prediction Markets

Quantitative Trader

New Haven, Connecticut

Apr 2026 – Present

- Engineered a Rust trading system on AWS that combines systematic market making and signal-based trading across Kalshi crypto and commodity markets, trading 200k+ contracts and earning \$1,200 per week on \$50/month of cloud infrastructure.
- Developed a sportsbook that answers live request-for-quote auctions for same-game parlays on Kalshi at up to 1,000 quotes per second, pricing correlated legs across NBA, NFL, MLB, and World Cup matches.

Yale University

Pre-Doctoral Researcher, Empirical Asset Pricing and Financial Econometrics

New Haven, Connecticut

Aug 2025 – Present

Advisor: Professor Bryan Kelly, Head of Machine Learning at AQR

- Demonstrated that ridge regression forecasts on news signals earn 9.7% annual alpha ($t=3.0$) on 25 commodity markets, after transaction costs and controls for momentum, value, basis, and basis-momentum.
- Published [Global Macroeconomic Data](#), a Python package that streams 1.5 TB of tick and macro data from LSEG Tick History, Datastream, WRDS, Compustat, OECD, and FRED through Polars into returns for 200+ global futures markets.
- Built an LLM pipeline that turns three decades of Dow Jones Newswires and Thomson Reuters articles into a dataset linking each article to its relevant futures markets and the paragraphs supporting each, running 30 million inference calls on Gemma 4 26B-A4B and Qwen 3.8-27B through vLLM.
- Cut GPU costs and wall time by 5x on Yale's SLURM high-performance computing cluster using prefill cache optimizations and keyword filtering before inference.

Amazon Web Services

Software Development Engineering Intern

Seattle, Washington

Jun 2025 – Aug 2025

- Shipped video performance metrics, including 7-day click rates and average watch time, to the Video Shopping Experiences portal used by over 2 million sellers and vendors, built end to end from SQL to Java to a localized React frontend.
- Recommended migrating the portal's metrics store from Elasticsearch to AWS Timestream to meet its latency and cost targets, working with the Timestream team and presenting to principal engineers and management, who adopted it.

University of Maryland, College Park

Research Assistant, Empirical Asset Pricing

College Park, Maryland

Oct 2023 – May 2025

Advisor: Professor Serhiy Kozak

- Modeled nonlinearities between firm characteristics with a kernel trick built in C++, CUDA, and Jax, achieving an annualized out-of-sample Sharpe ratio of 3.2 against 1.65 for PCA on 40 anomaly portfolios.
- Extended factor models with a lag dimension, using past characteristics to model the term structure of characteristic portfolio returns for multi-horizon investing, cutting out-of-sample pricing errors by 10 to 18% against a pooled PCA factor model.

Capital One

Software Engineering Intern

McLean, Virginia

Jun 2024 – Aug 2024

Northrop Grumman

Software Engineering Intern

Boulder, Colorado

Jun 2023 – Aug 2023

SKILLS

Languages: Python, Rust, Java, C++, CUDA, SQL, JavaScript

Libraries: PyTorch, Scikit-learn, Statsmodels, Jax, vLLM, Transformers, Pandas, NumPy, Polars, React, FastAPI

Tools: AWS (S3, Lambda, EC2, Lightsail), SLURM, Hugging Face, Elasticsearch, Kalshi API, WebSockets, Claude Code